Multiple regression part 2: curves and interactions

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Data for today

```
dat <- read_csv("https://www.openintro.org/data/csv/acs12.csv")
glimpse(dat)
```

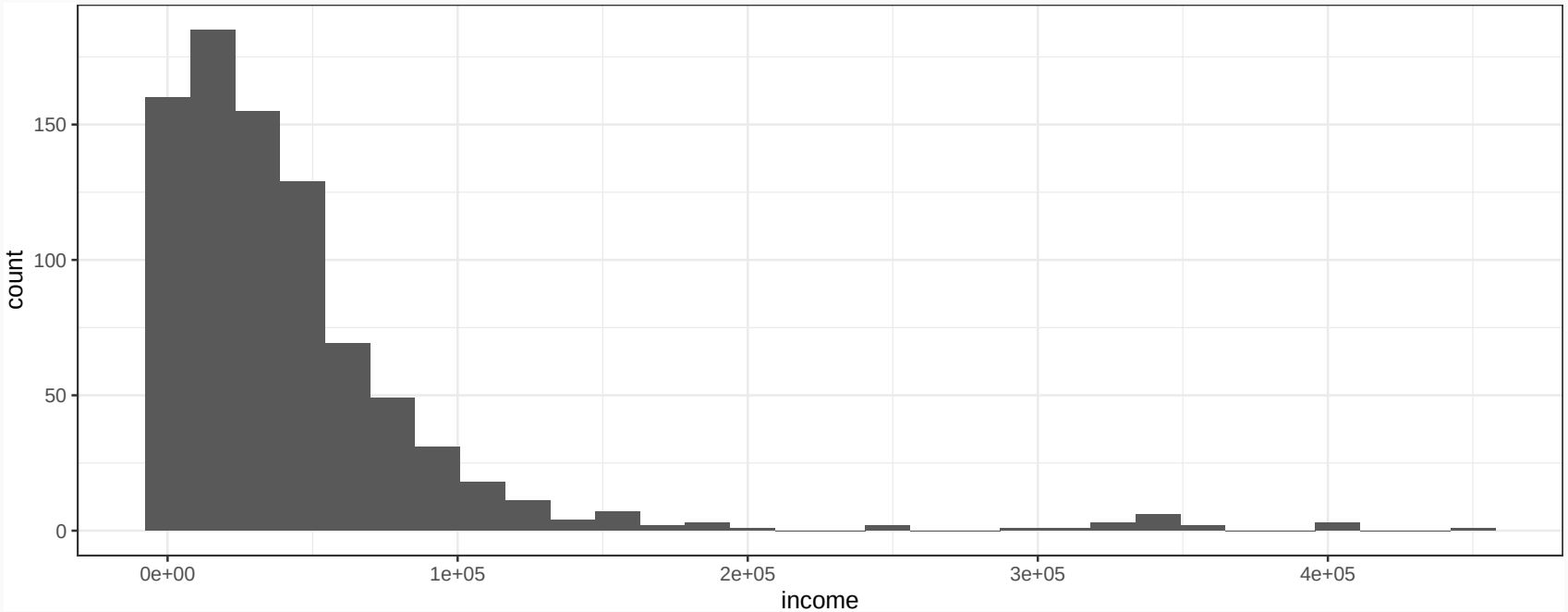
```
Rows: 2,000
Columns: 13
$ income       <dbl> 60000, 0, NA, 0, 0, 1700, NA, NA, NA, 45000, NA, 8600, 0,~
$ employment   <chr> "not in labor force", "not in labor force", NA, "not in l~
$ hrs_work     <dbl> 40, NA, NA, NA, NA, 40, NA, NA, NA, 84, NA, 23, NA, NA, N~
$ race         <chr> "white", "white", "white", "white", "white", "other", "wh~
$ age          <dbl> 68, 88, 12, 17, 77, 35, 11, 7, 6, 27, 8, 69, 69, 17, 10, ~
$ gender       <chr> "female", "male", "female", "male", "female", "female", "~
$ citizen      <chr> "yes", "yes", "yes", "yes", "yes", "yes", "yes", "yes", "~
$ time_to_work <dbl> NA, NA, NA, NA, NA, 15, NA, NA, NA, 40, NA, 5, NA, NA, NA~
$ lang         <chr> "english", "english", "english", "other", "other", "other~
$ married      <chr> "no", "no", "no", "no", "no", "yes", "no", "no", "no", "y~
$ edu          <chr> "college", "hs or lower", "hs or lower", "hs or lower", "~
$ disability   <chr> "no", "yes", "no", "no", "yes", "yes", "no", "yes", "no",~
$ birth_qrtr   <chr> "jul thru sep", "jan thru mar", "oct thru dec", "oct thru~
```

Subset the data to only include workers

```
dat <- dat |>
    filter(employment == "employed")
```

Let’s look at income for this ACS 2012 sample

```
ggplot(dat, aes(x = income)) + geom_histogram()
```



Multiple regression (regression with more than 1 predictor)

We can generalize the linear regression

$$Y = \beta_0 + \beta_1 X + \varepsilon$$
$$\varepsilon \sim N(0, \sigma^2)$$

Multiple regression (regression with more than 1 predictor)

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as

$$Y = \beta_0 + \beta_1 x_1 \cdots \beta_k x_k + \varepsilon$$
$$\varepsilon \sim N(0, \sigma^2)$$

Where k is the number of predictor variables we include in the model. Our only constraint is that k must be smaller than the number of observations n in our data.

Multiple regression basics

1. Categorical predictors act as intercepts, or differences in level
2. Continuous predictors act as slopes

Our model for income

Our theoretical model tells us that if we want to learn about $t \longrightarrow i$, we must adjust for the effects that e has on both t and i.

```
m1 <- lm(income ~ hrs_work + edu, data = dat)
```

Interpreting this model

```
table(dat$edu)
```

```
    college        grad hs or lower
        241            99         503
```

```
tidy(m1)
```

```
# A tibble: 4 x 5
  term           estimate std.error statistic  p.value
  <chr>             <dbl>     <dbl>     <dbl>    <dbl>
1 (Intercept)        166.     6248.    0.0266 9.79e- 1
2 hrs_work          1279.      135.    9.49   2.30e-20
3 edugrad          44423.     5997.    7.41   3.15e-13
4 eduhs or lower  -18591.     3935.   -4.72   2.70e- 6
```

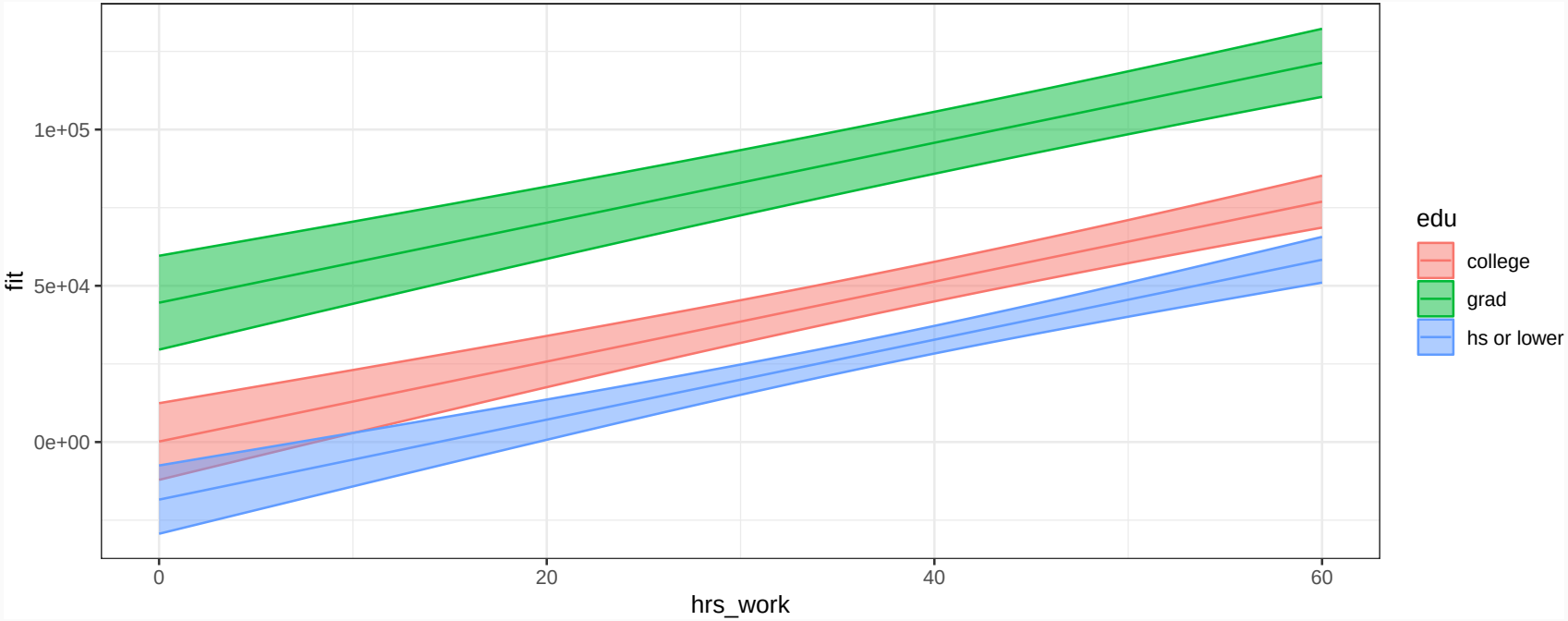
To understand what the model says, let's visualize y-hat

```
# set up prediction data with values of interest
hrs_work <- 0:60
edu <- c("college", "hs or lower", "grad")
pred_dat <- expand_grid(hrs_work, edu)
# generate expected values and CI, join pred_dat
e_y <- predict(m1, newdata = pred_dat, interval = "confidence") |>
    bind_cols(pred_dat)
# inspect
head(e_y)
```

```
# A tibble: 6 x 5
      fit     lwr    upr hrs_work edu
    <dbl>   <dbl>  <dbl>    <int> <chr>
1    166. -12098. 12430.        0 college
2 -18425. -29351. -7499.        0 hs or lower
3  44589.  29568. 59610.        0 grad
4   1445. -10593. 13484.        1 college
5 -17145. -27829. -6461.        1 hs or lower
6  45868.  31046. 60691.        1 grad
```

Visualizing model expectations

```
ggplot(e_y, aes(y = fit, ymin = lwr, ymax = upr, x = hrs_work, fill = edu, color = edu)) +
    geom_ribbon(alpha = 0.5) + geom_line()
```



Regression with more than one slope and multiple intercepts

Let's see if income depends on three variables: education, income, and hours worked

$$E(\text{income}) = \beta_0 + \beta_1 \text{hrs} + \beta_2 \text{edu} + \beta_3 \text{age}$$

Fitting the model

```r
m2 <- lm(income ~ hrs_work + edu + age, data = dat)

broom::tidy(m2)
```

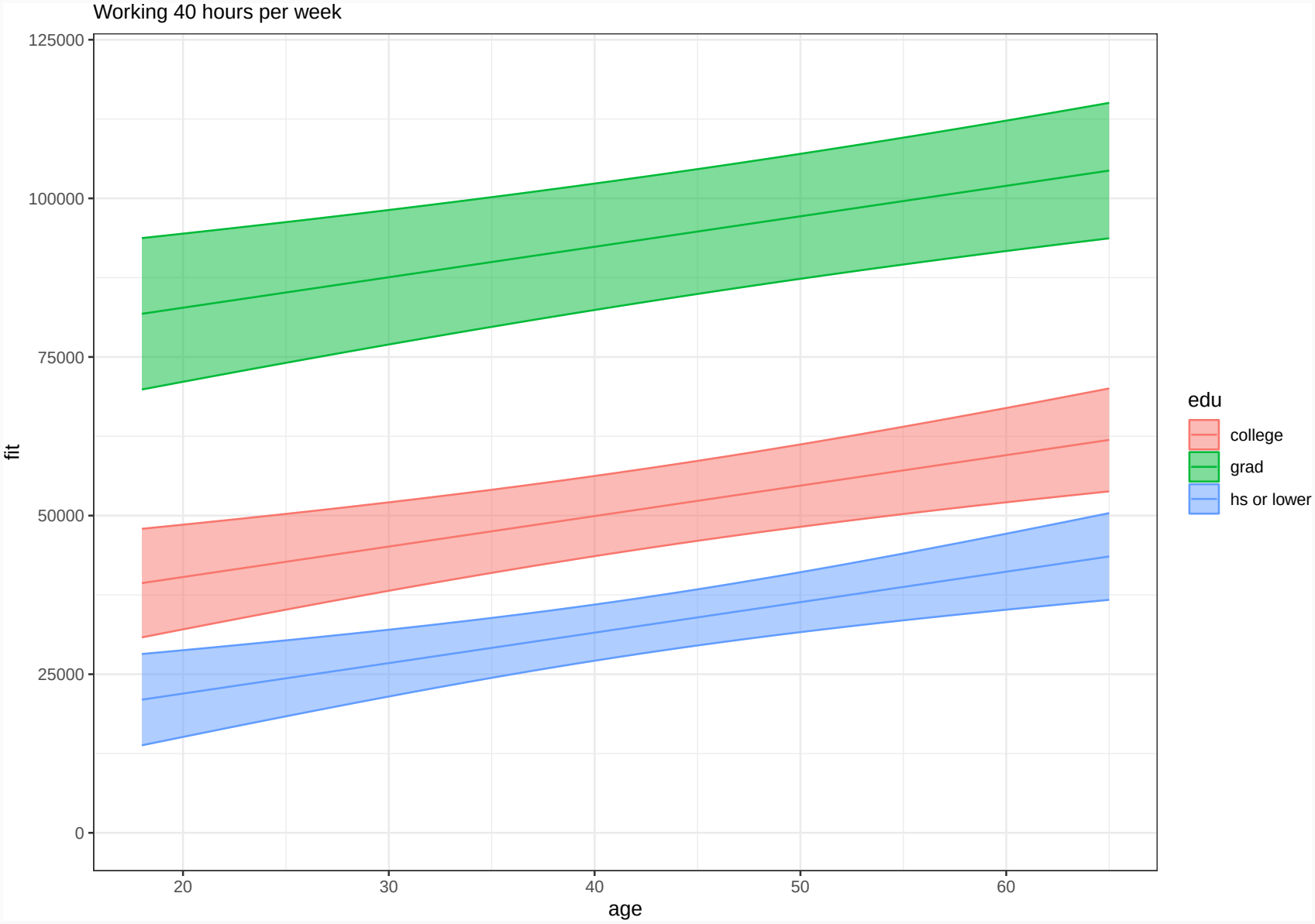
```
# A tibble: 5 x 5
  term           estimate std.error statistic  p.value
  <chr>             <dbl>     <dbl>     <dbl>    <dbl>
1 (Intercept)     -18761.     7761.     -2.42 1.58e- 2
2 hrs_work          1237.      134.      9.23 2.14e-19
3 edugrad          42446.     5963.      7.12 2.36e-12
4 eduhs or lower  -18372.     3899.     -4.71 2.88e- 6
5 age                480.      119.      4.04 5.72e- 5
```

Visualizing model expectations: setup

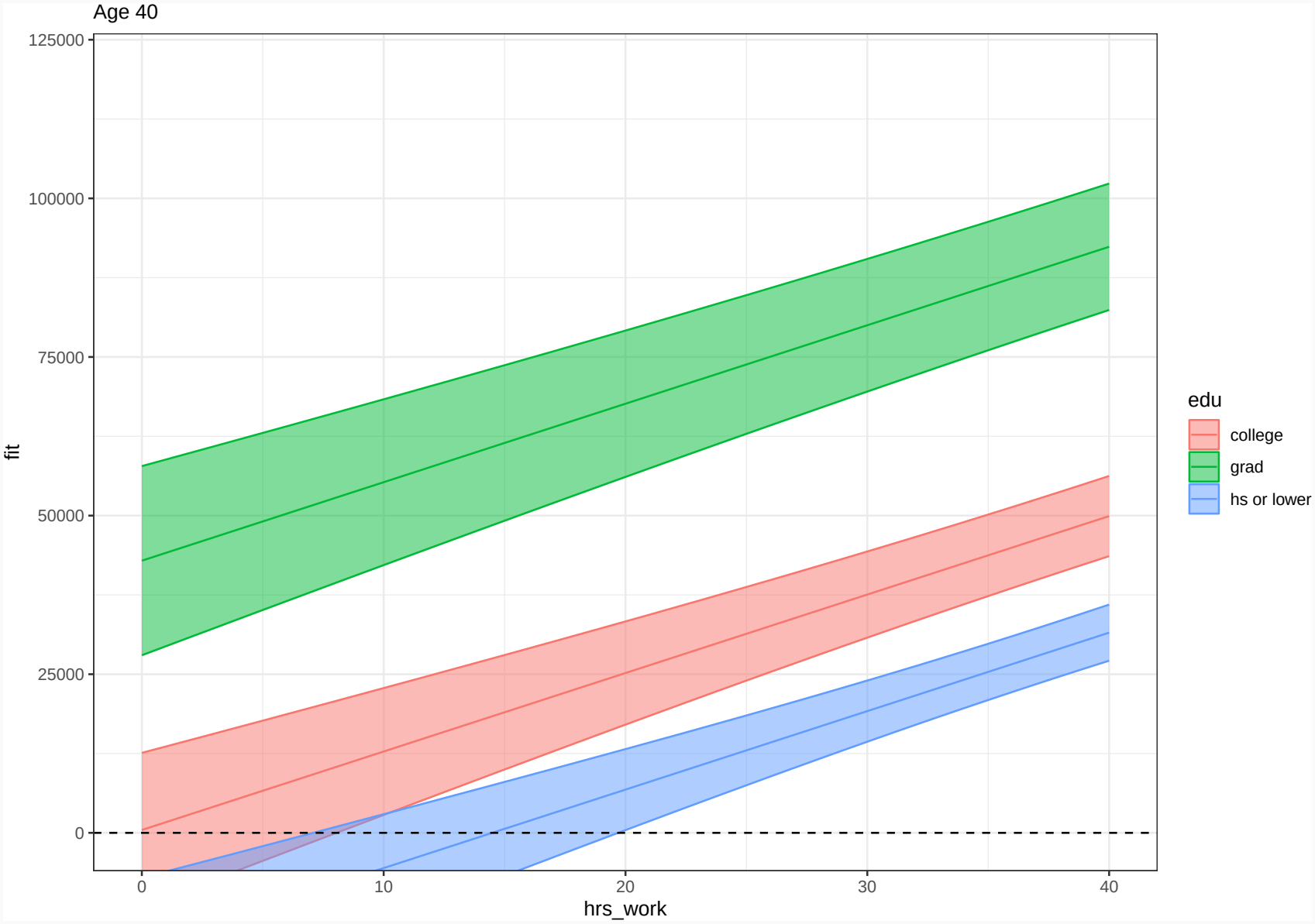
```
# set up prediction data with values of interest
hrs_work <- 0:40
age <- c(18:65)
edu <- c("college", "hs or lower", "grad")
pred_dat <- expand_grid(hrs_work, edu, age)
# generate expected values and CI, join pred_dat
e_y <- predict(m2, newdata = pred_dat, interval = "confidence") |>
    bind_cols(pred_dat)
# inspect
head(e_y)
```

```
# A tibble: 6 x 6
      fit     lwr   upr hrs_work edu       age
    <dbl>   <dbl> <dbl>    <int> <chr>   <int>
1 -10118. -23255. 3020.        0 college    18
2  -9637. -22689. 3414.        0 college    19
3  -9157. -22125. 3811.        0 college    20
4  -8677. -21566. 4212.        0 college    21
5  -8197. -21010. 4616.        0 college    22
6  -7717. -20458. 5024.        0 college    23
```

Visualizing model expectations: age and expected income



Hours worked and expected income



Adding complexity: quadratic terms

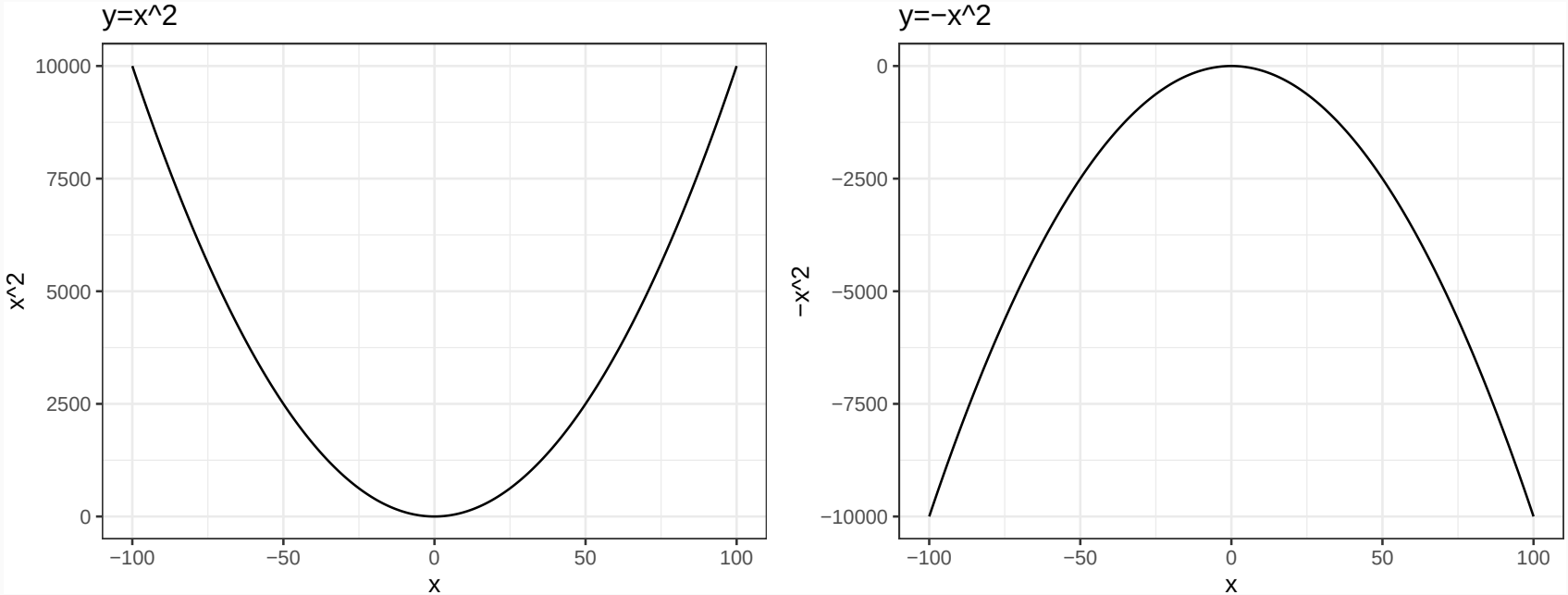
We know that earnings for people less than age 18 and greater than age 70 tend to be very low (or zero). We can try to use a parabola (a quadratic equation) to model this process.

Our revised model will be

$$E(\text{income}) = \beta_0 + \beta_1 \text{hrs} + \beta_2 \text{edu} + \beta_3 \text{age} + \beta_4 \text{age}^2$$

Quadratics take a form that looks like this

```
pdat <- data.frame(x = -100:100)
p1 <- ggplot(pdat, aes(x = x, y = x^2)) + geom_line() + labs(title = "y=x^2")
p2 <- ggplot(pdat, aes(x = x, y = -x^2)) + geom_line() + labs(title = "y=-x^2")
p1 + p2
```



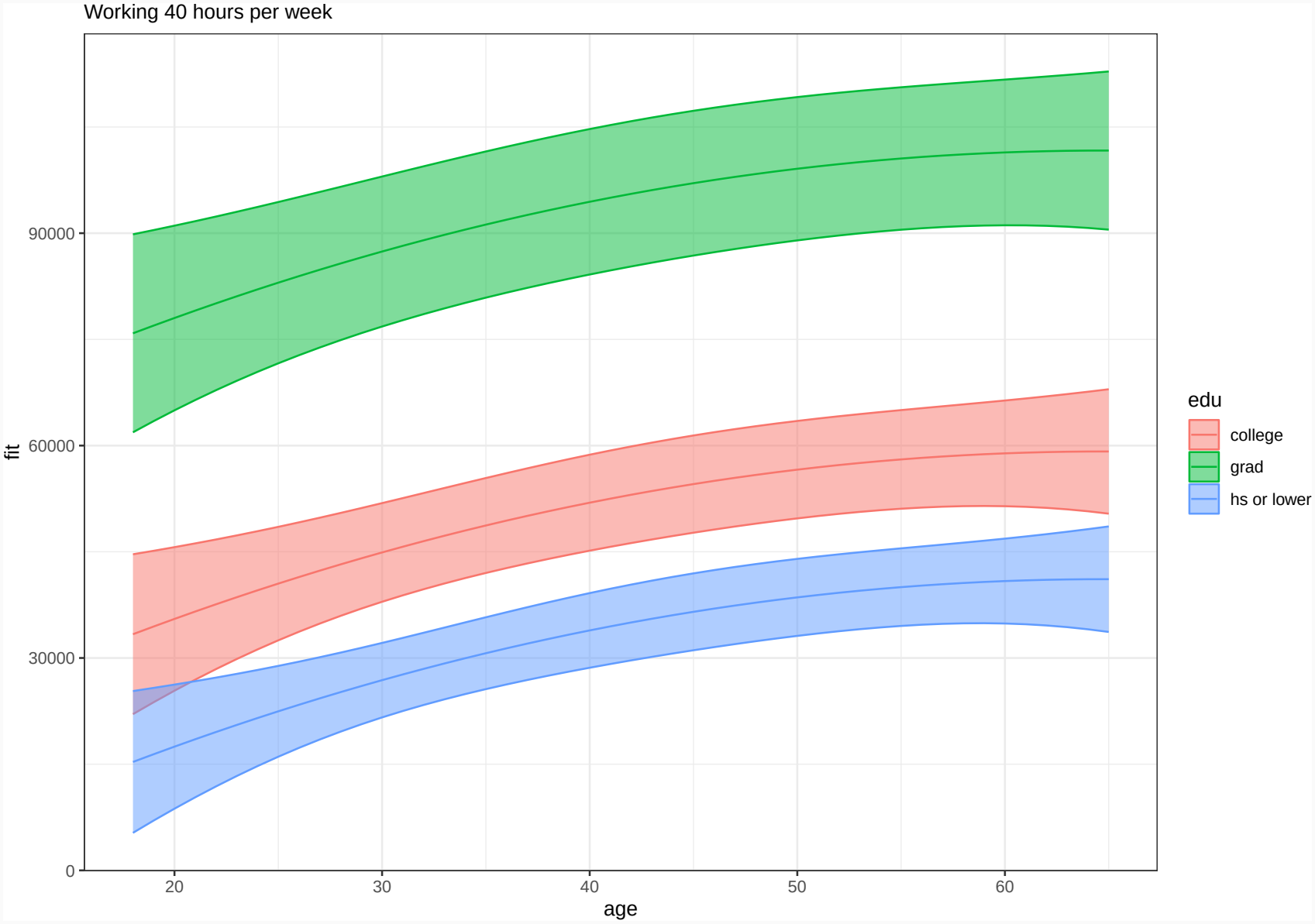
Fitting a quadratic term

We use the I() function to require R to evaluate math statements inside formula objects

```
m3 <- lm(income ~ hrs_work + edu + age + I(age^2), data = dat)
tidy(m3)
```

```
# A tibble: 6 x 5
  term           estimate std.error statistic  p.value
  <chr>             <dbl>     <dbl>     <dbl>    <dbl>
1 (Intercept)    -37114.   13856.       -2.68 7.54e- 3
2 hrs_work         1169.     140.        8.33 3.40e-16
3 edugrad         42503.    5958.        7.13 2.11e-12
4 eduhs or lower -18040.    3901.       -4.62 4.36e- 6
5 age              1529.     667.        2.29 2.21e- 2
6 I(age^2)          -11.8      7.39     -1.60 1.10e- 1
```

Visualizing model expectations



What about gender and education?

Perhaps we think that gender influences income. We can expand our model by including yet another predictor.

```
m4<-lm(income ~ -1 + # remove b0 to aid interpretation
         hrs_work +
         edu +
         age +
         gender,
       data = dat)
```

What does this model say?

```
summary(m4)
```

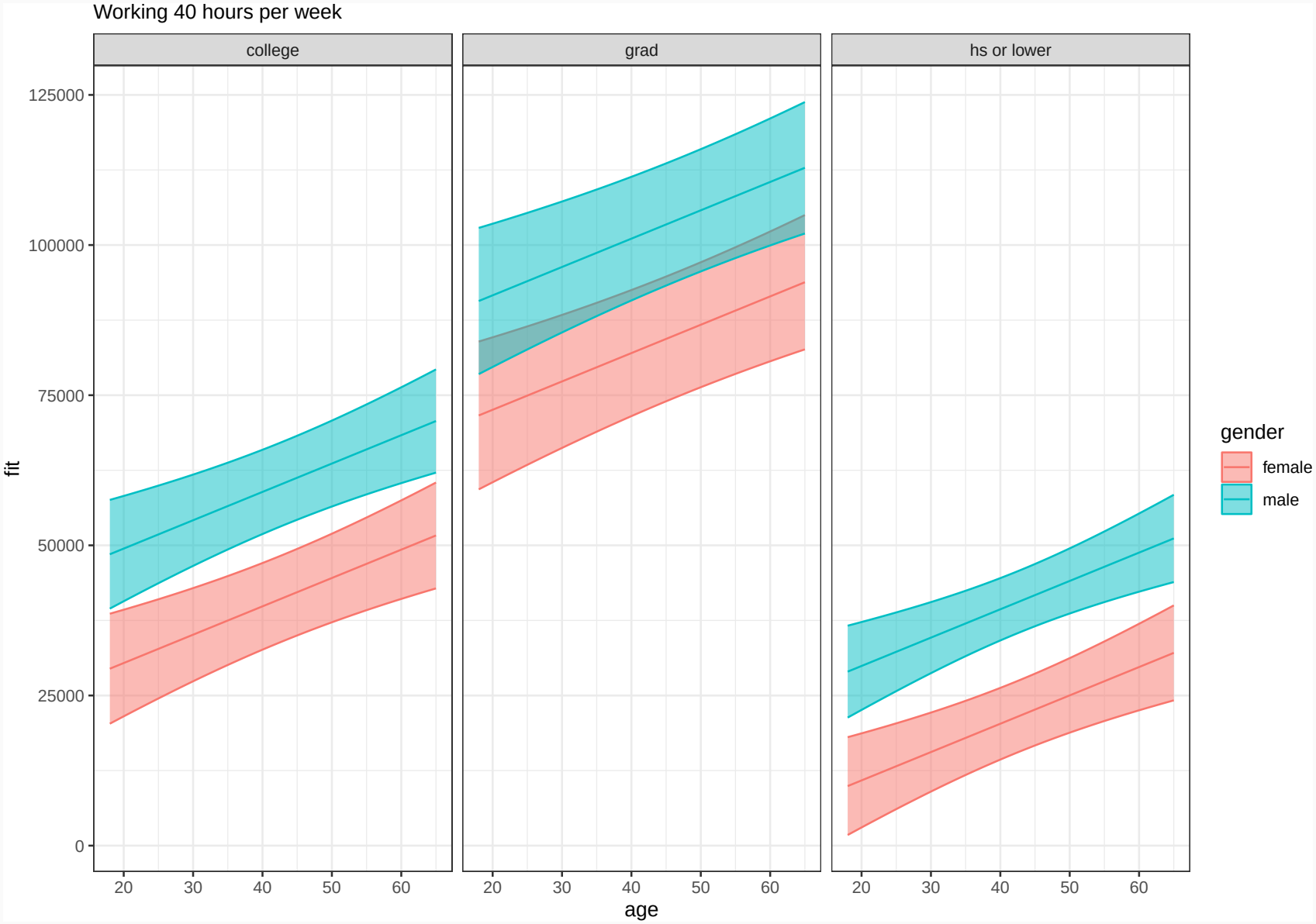
```
Call:
lm(formula = income ~ -1 + hrs_work + edu + age + gender, data = dat)

Residuals:
    Min      1Q  Median      3Q     Max
-124612  -20192   -4229   10060  321601

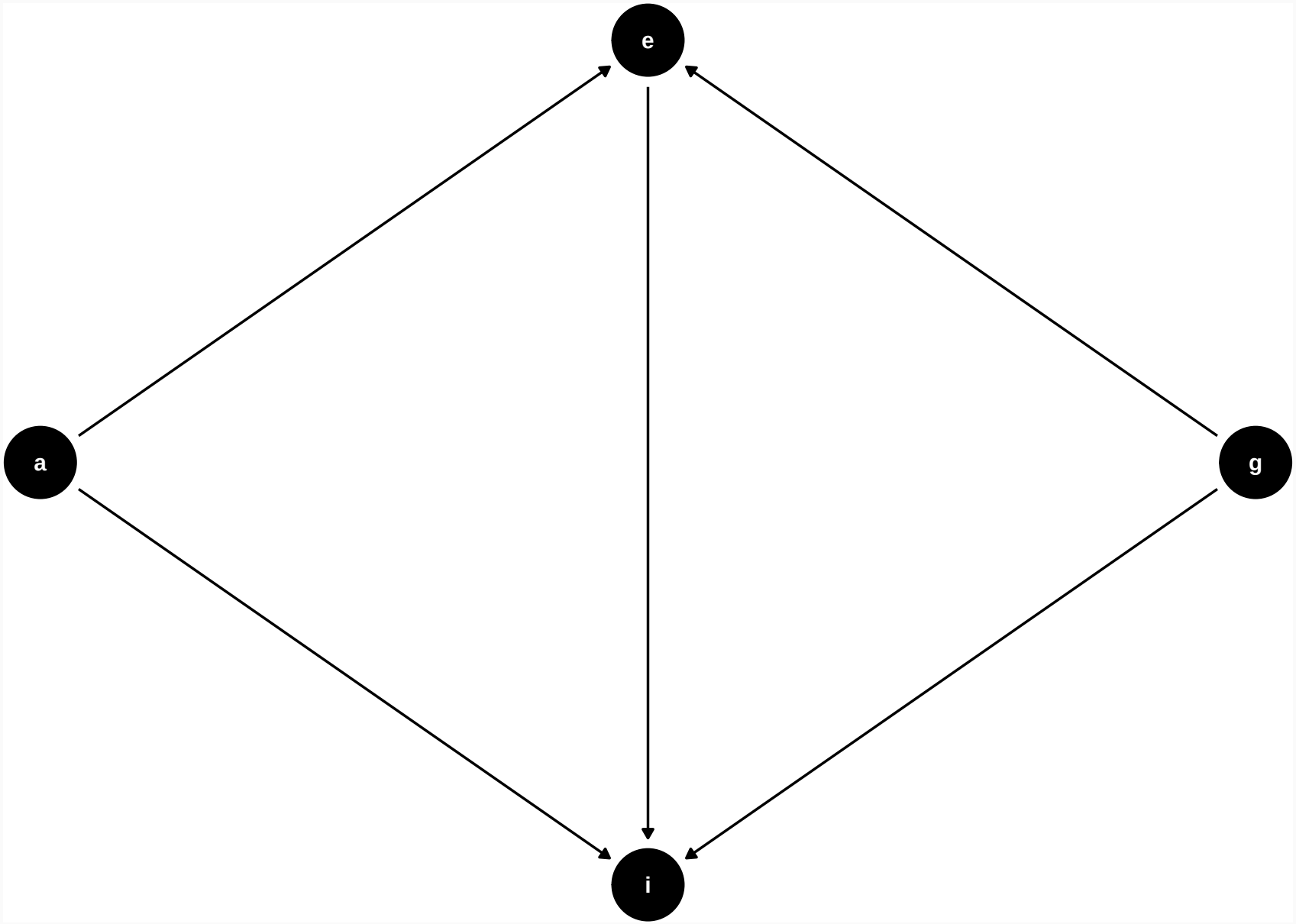
Coefficients:
               Estimate Std. Error t value Pr(>|t|)
hrs_work         1035.8      136.9   7.564 1.03e-13 ***
educollege     -20467.6     7640.2  -2.679  0.00753 **
edugrad         21698.5     9034.0   2.402  0.01653 *
eduhs or lower -40012.1     7100.5  -5.635 2.39e-08 ***
age               471.9      116.8   4.041 5.81e-05 ***
gendermale      19054.6     3524.7   5.406 8.41e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 48850 on 837 degrees of freedom
Multiple R-squared:  0.5446,    Adjusted R-squared:  0.5413
F-statistic: 166.8 on 6 and 837 DF,  p-value: < 2.2e-16
```

Model expected values



What if we think the impact of education on income depends on gender?



Allowing the g->i relationship to vary by education

We can fit an *interaction* to the model to allow for separate parameters for all education and gender pairs

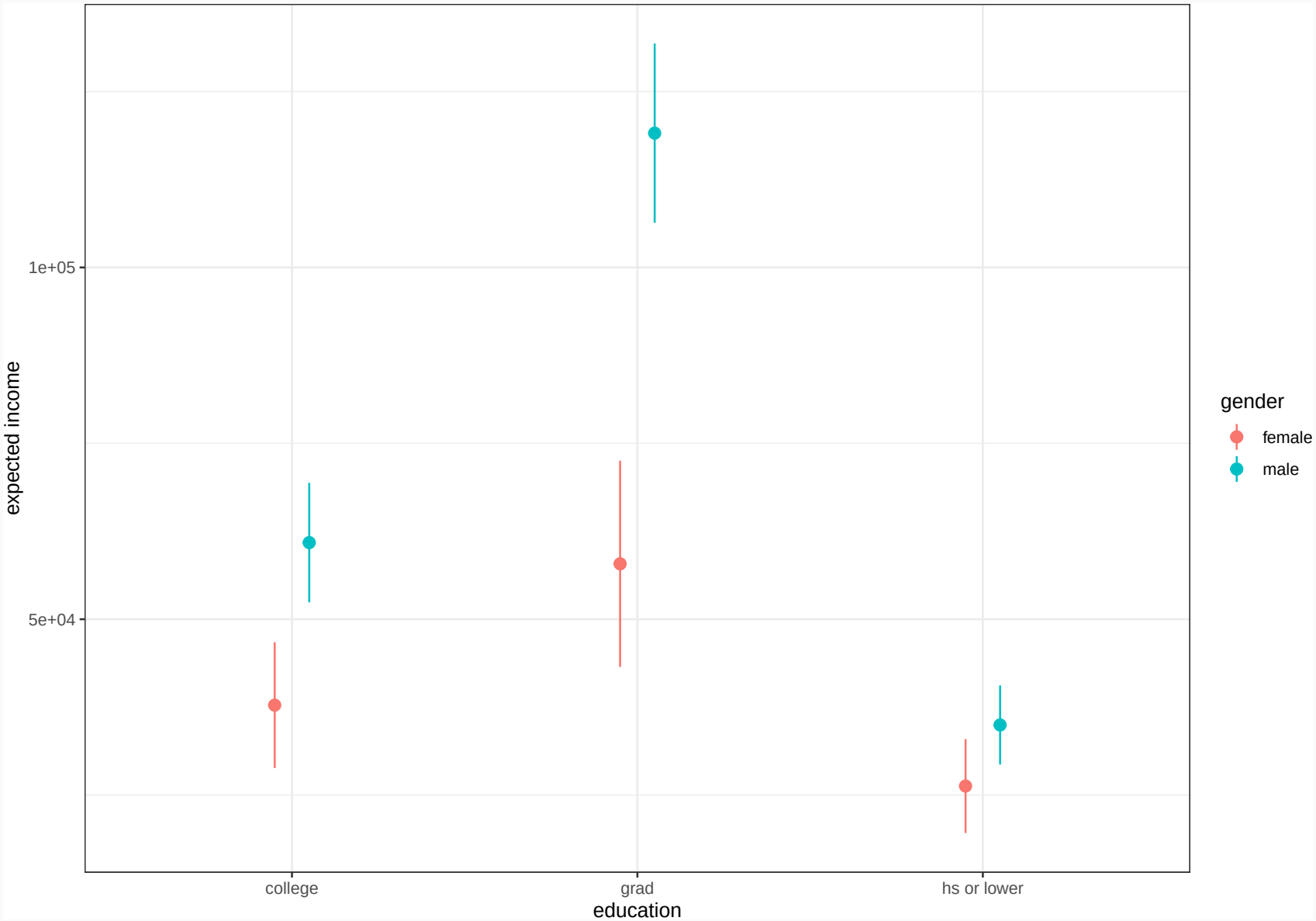
```
m5<-lm(income ~ -1 + # remove b0 to aid interpretation
         hrs_work +
         age +
         edu * gender, # interaction
       data = dat)
```

Interpreting model output: so many parameters!

```
broom::tidy(m5)
```

```
# A tibble: 8 x 5
  term                      estimate std.error statistic  p.value
  <chr>                        <dbl>     <dbl>     <dbl>    <dbl>
1 hrs_work                     1051.      135.     7.78  2.18e-14
2 age                           447.      115.     3.87  1.18e- 4
3 educollege                 -22125.     7951.    -2.78  5.51e- 3
4 edugrad                     -2037.    10372.    -0.196 8.44e- 1
5 eduhs or lower             -33625.     7236.    -4.65  3.91e- 6
6 gendermale                  23103.     6301.     3.67  2.62e- 4
7 edugrad:gendermale          38114.    11612.     3.28  1.07e- 3
8 eduhs or lower:gendermale  -14423.     7597.    -1.90  5.80e- 2
```

Visualizing the interaction



Categorical x categorical interactions

Interacting two categorical variables generates intercepts for each combination of levels of the two variables. Interacting a 2 level categorical with a 3 level categorical generates $2 \times 3 = 6$ intercepts.

Categorical x continuous interactions

Let's allow the relation between age and income to vary by sex.

$$E(\text{income}) = \beta_0 + \beta_1 \text{hrs} + \beta_2 \text{edu} + \beta_3 \text{age} + \beta_4 \text{gender} + \beta_5 \text{gender x age}$$

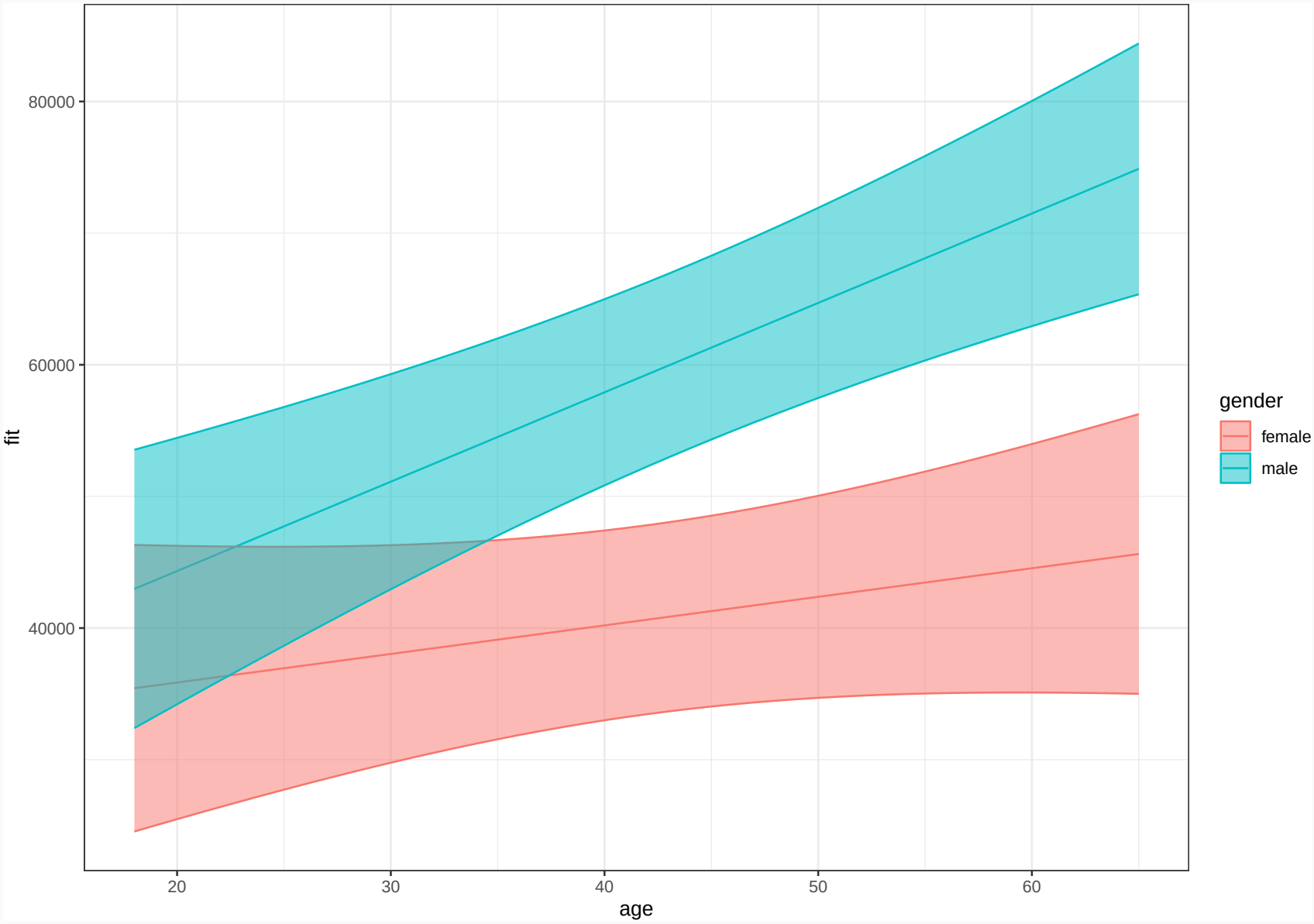
Fitting the model: interpretation is tricky

```r
m6<-lm(income ~ hrs_work + edu +
         age * gender, # interaction
       data = dat)

broom::tidy(m6)
```

```
# A tibble: 7 x 5
  term           estimate std.error statistic  p.value
  <chr>             <dbl>     <dbl>     <dbl>    <dbl>
1 (Intercept)      -9737.     9353.   -1.04   2.98e- 1
2 hrs_work          1032.      137.    7.55   1.18e-13
3 edugrad          42200.     5855.    7.21   1.28e-12
4 eduhs or lower  -19109.     3841.   -4.97   7.93e- 7
5 age                217.      174.    1.25   2.12e- 1
6 gendermale        -774.    10605.   -0.0730 9.42e- 1
7 age:gendermale     462.      233.    1.98   4.78e- 2
```

But visualization makes it clear



Categorical x continuous interactions

Interacting a categorical and a continuous variable generates different slopes for the continuous measure for each level of the categorical variable.

Another example

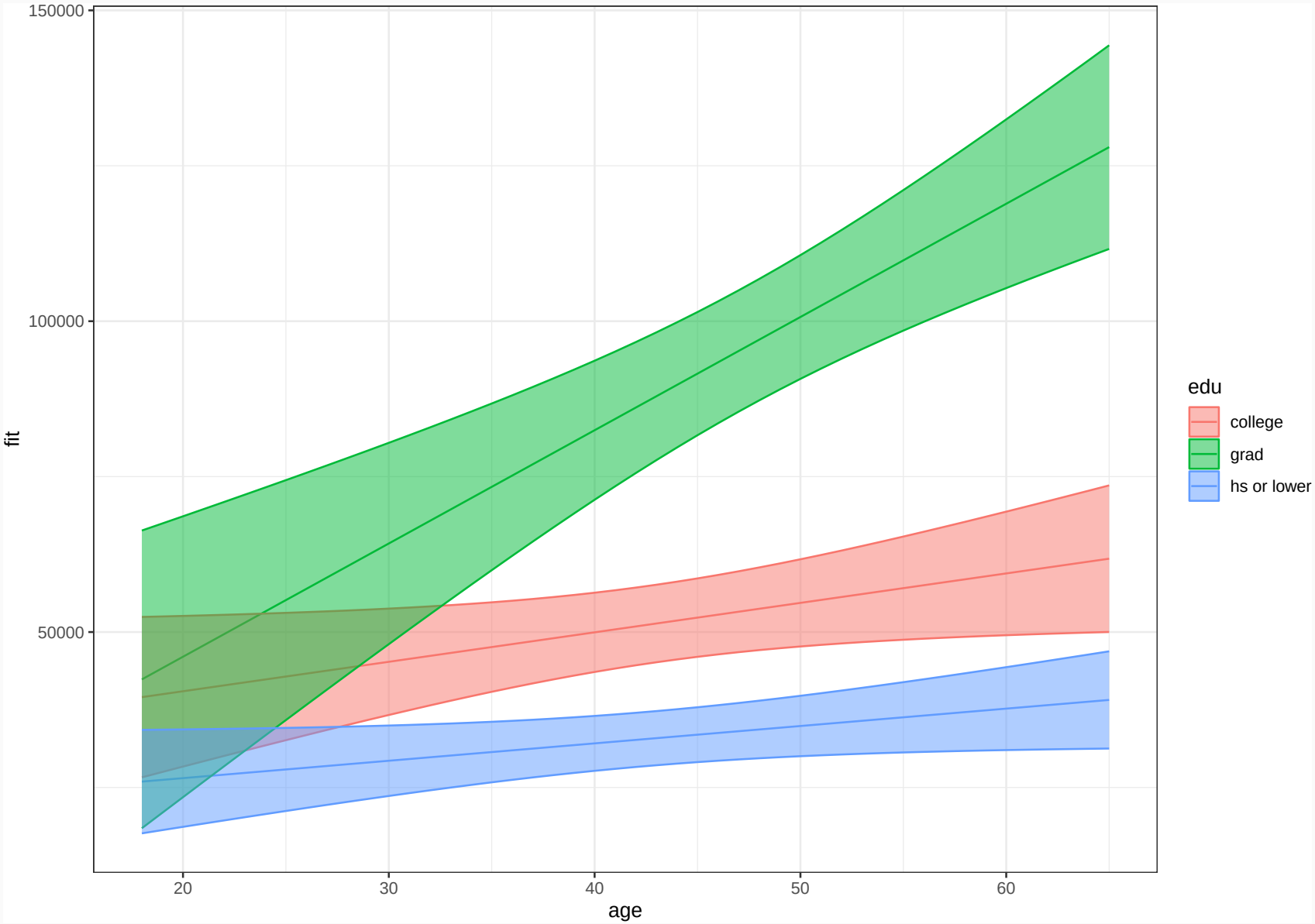
Let's allow for differences in the slope of age by education

```
m7<-lm(income ~ hrs_work + edu * age, # interaction
       data = dat)

broom::tidy(m7)
```

```
# A tibble: 7 x 5
  term               estimate std.error statistic  p.value
  <chr>                 <dbl>     <dbl>     <dbl>    <dbl>
1 (Intercept)         -19915.    11729.    -1.70  8.99e- 2
2 hrs_work              1273.      134.     9.53  1.69e-20
3 edugrad             -21404.    21352.    -1.00  3.16e- 1
4 eduhs or lower      -10089.    12353.    -0.817 4.14e- 1
5 age                    474.      231.     2.05  4.04e- 2
6 edugrad:age           1348.      444.     3.03  2.49e- 3
7 eduhs or lower:age    -194.      274.    -0.709 4.78e- 1
```

Visualize


150000
100000
50000
fit
20
30
40
50
60
age
edu
college
grad
hs or lower

Continuous x continuous interactions

What if we think the effect of hours worked depends on age?

$$E(\text{income}) = \beta_0 + \beta_1 \text{hrs} + \beta_2 \text{age} + \beta_3 \text{age} \times \text{hrs}$$

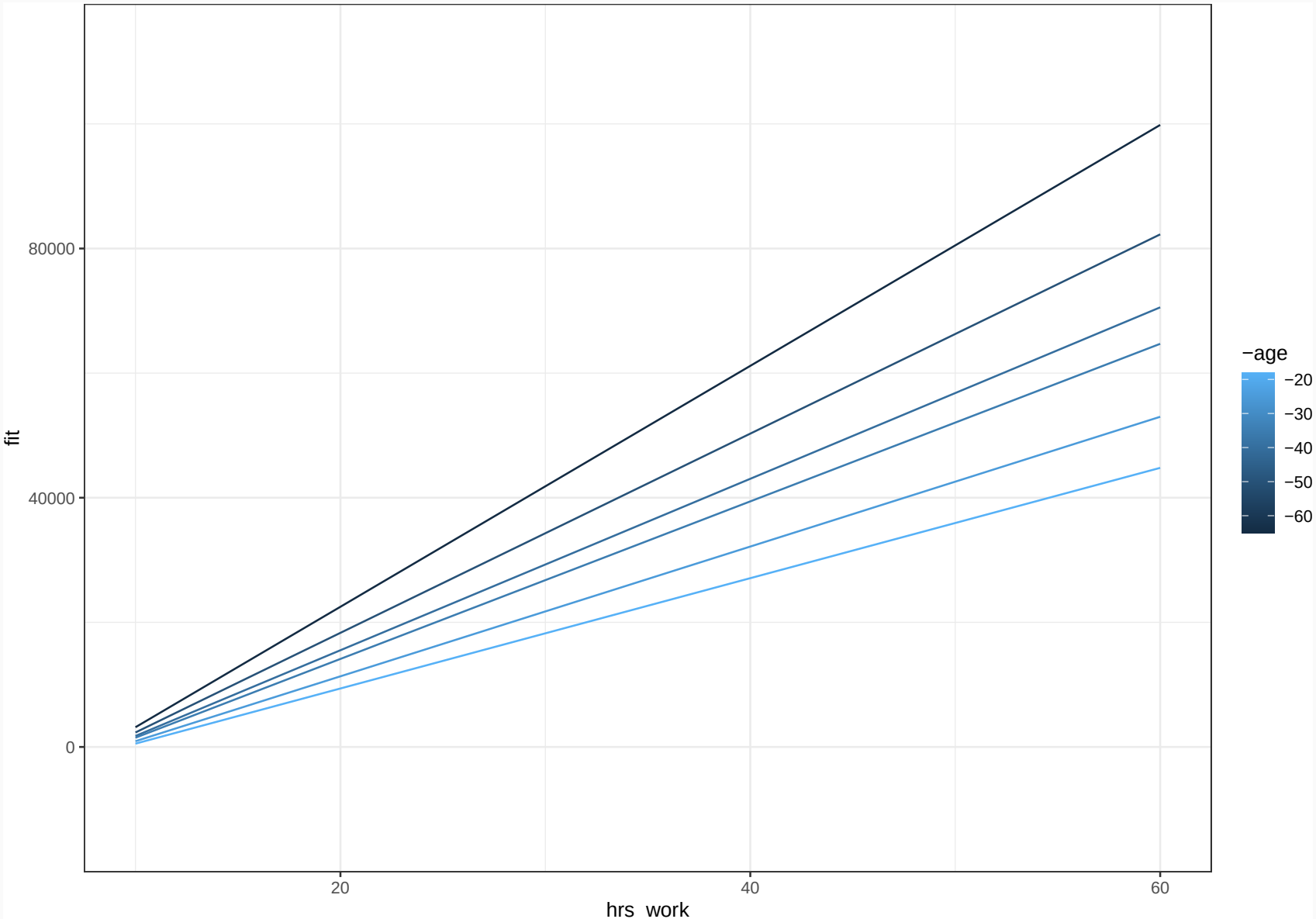
```
m8<-lm(income ~ hrs_work * age, # interaction
       data = dat)

broom::tidy(m8)
```

```
# A tibble: 4 x 5
  term         estimate std.error statistic p.value
  <chr>           <dbl>     <dbl>     <dbl>   <dbl>
1 (Intercept)   -5316.   14788.      -0.359  0.719
2 hrs_work        484.     398.       1.22   0.224
3 age            -167.     335.      -0.499  0.618
4 hrs_work:age     22.3      8.94     2.50   0.0128
```

Visualization

These are really tricky to interpret, you'll need to visualize it



Continuous x continuous interactions

Create constant changes in slope as both measures change. The allow for a very flexible model fit to the data, but are difficult to interpret directly.

Combining interactions

Sometimes it makes sense to *fully* interact your model, allow each measure to interact with all other measures. Let's see this with age, hrs_work, and education

```
m9<-lm(income ~ hrs_work * age * edu, # interaction
       data = dat)
broom::tidy(m9)
```

```
# A tibble: 12 x 5
   term                        estimate std.error statistic p.value
   <chr>                          <dbl>     <dbl>     <dbl>   <dbl>
 1 (Intercept)                   -9415.    26153.    -0.360    0.719
 2 hrs_work                        888.      681.     1.30     0.193
 3 age                            -224.      581.    -0.386    0.700
 4 edugrad                       92523.    85645.     1.08     0.280
 5 eduhs or lower                 1295.    31012.     0.0418   0.967
 6 hrs_work:age                     20.4      15.4    1.32     0.186
 7 hrs_work:edugrad              -2644.     1977.    -1.34     0.181
 8 hrs_work:eduhs or lower        -286.      826.    -0.346    0.729
 9 age:edugrad                   -1088.     1604.    -0.679    0.498
10 age:eduhs or lower              306.      701.     0.437    0.663
11 hrs_work:age:edugrad             56.0      37.6    1.49     0.137
12 hrs_work:age:eduhs or lower     -13.2      18.8   -0.700    0.484
```

Visualization

What looks off here?

